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Network slicing group

Abstract

Embodiments provide novel approaches to handling network slice groups in access stratum and non-access stratum communications in a cellular communication network. Novel slice identification schema can support group-level identification of groups of network slices without individually listing the network slices. Some embodiments implement a slice identifier that supports wildcard slice/service type (SST) options and/or wildcard slice differentiator (SD) options to support wildcard-enabled slice groups. Some embodiments enable use of such wildcard-enabled slice groups in context of unified access control. Some embodiments introduce a novel criteria type to unified access control to expressly support flexible slice grouping.

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Background/Summary

FIELD

(1) Embodiments relate generally to wireless communication networks, and, more particularly to group-level identification and exploitation of network slice groups.

BACKGROUND

(2) Some modern wireless communication networks, such as fifth generation (5G) wireless networks, have begun to support so-called network slicing. The communication network includes physical (e.g., multi-domain) infrastructure, such as transport network infrastructure, core network infrastructure, and access network infrastructure. Virtual network operators (VNOs), also referred to as mobile network operators (MNOs), mobile virtual network operators (MVNOs), and the like can concurrently use the physical infrastructure to offer multiple virtualized networks.

(3) Such network slices can be implemented using various virtualization technologies, such as software-defined networking (SDN) and network functions virtualization (NFV) (also referred to as virtualized network functions (VNFs)). Each of those virtualized networks is considered to be a network slice, and each network slice can be designed (e.g., optimized) to support a particular desired set of users, applications, services, network functions, etc. Each network slice can be implemented with its own management domain and its own automation, and each network slice can be logically separated from other network slices. For example, different network slices can be used to separate and/or prioritize traffic, manage security, maintain separation between customers and/or services, etc.

(4) While the concept of network slicing was originally to enable self-contained virtualized networks, some VNOs have more recently found use cases for grouping network slices. For example, a VNO may offer multiple network slice types (e.g., optimized for particular categories of services) to each of multiple customers. When a particular user equipment (UE) device sends an attachment request to connect with the wireless network via an eNodeB, or other suitable network node, the VNO may attach the UE to the network concurrently via multiple network slices by associating the UE's communications with an explicit list of the identifier of each network slice that is part of the group. Particularly in cases where it is desirable to attach the UE to a large number of network slices, such as to all networks slices for a particular customer, such explicit listing of network slice identifiers can be inefficient.

SUMMARY

(5) Embodiments provide novel approaches to handling network slice groups in access stratum and non-access stratum communications in a cellular communication network. Novel slice identification schema can support group-level identification of groups of network slices without individually listing the network slices. Some embodiments implement a slice identifier that supports wildcard slice/service type (SST) options and/or wildcard slice differentiator (SD) options to support wildcard-enabled slice groups. Some embodiments enable use of such wildcard-enabled slice groups in context of unified access control. Some embodiments introduce a novel criteria type to unified access control to expressly support flexible slice grouping.

(6) According to one set of embodiments, a method is provided for network slice grouping in a cellular communication network infrastructure. The method includes: receiving a user equipment device (UE) message by one or more network function (NF) nodes of the cellular communication network infrastructure, the UE message being received from a UE of a plurality of UE devices via a radio access network (RAN); determining, by the at least one NF nodes, that the UE message invokes a wildcard-enabled slice group that is subset of a plurality of network slices, each network slice being a respective one of a plurality of virtualized data subnetworks, each virtualized data subnetwork uniquely identifiable by a respective one of a plurality of network slice identifiers, each network slice identifier having a respective unique combination of a slice/service type (SST) value and a slice differentiator (SD) value; generating, by the at least one NF nodes in response to the UE message, a NF message that identifies the wildcard-enabled slice group uniquely by one of the

plurality of network slice identifiers that has either the SST value set to a predefined wildcard SST value or has the SD value set to a predefined wildcard SD value; and communicating the NF message to the UE via the RAN.

(7) According to another set of embodiments, a system is provided for network slice grouping in a cellular communication network infrastructure. The system includes: a plurality of virtualized data subnetworks built on the cellular communication network infrastructure, each virtualized data subnetwork uniquely being one of a plurality of network slices identifiable by a respective one of a plurality of network slice identifiers, each network slice identifier having a respective unique combination of a slice/service type (SST) value and a slice differentiator (SD) value; and one or more network function (NF) nodes in communication with the plurality of virtualized data subnetworks and in communication with a plurality of user equipment devices (UEs) via one or more radio access networks (RANs). The one or more NF nodes include: one or more processors; and non-transitory memory having instructions stored thereon, which, when executed, cause the one or more processors to perform steps. The steps include: receiving a UE message from a UE of the plurality of UE devices via a RAN of the one or more RANs; determining that the UE message invokes a wildcard-enabled slice group that is subset of a plurality of network slices and is uniquely identifiable by one of the plurality of network slice identifiers that has either the SST value set to a predefined wildcard SST value or has the SD value set to a predefined wildcard SD value; and communicating, to the UE via the RAN, in response to the UE message, a NF message that includes the one of the plurality of network slice identifiers to uniquely identify the wildcard-enabled slice group.

(8) According to another set of embodiments, a method is provided for unified access control of network slice groups in a cellular communication network infrastructure. The method includes: negotiating, between an access and mobility management function (AMF) and a user equipment device (UE) via a radio access network (RAN), a unified access control schema corresponding to an operator-defined access category that indicates a criteria type configured to support slice grouping; mapping a slice group identifier to a corresponding value field of a barring information message in accordance with the unified access control schema, the slice group identifier uniquely identifying an operator-defined group of network slices of a plurality of network slices, each of the plurality of network slices being uniquely identifiable by a network slice identifier separate from the slice group identifier; and communicating the barring information message from the AMF to the UE via the RAN to indicate to the UE which of the plurality of network slices is barred from use in establishing communication sessions between the UE and the cellular communication network infrastructure.

(9) According to another set of embodiments, another method is provided for unified access control of network slice groups in a cellular communication network infrastructure. The method includes: receiving, by a user equipment device (UE) from an access and mobility management function (AMF) via a radio access network (RAN), a barring information message indicating to the UE which of a plurality of network slices is barred from use in establishing communication sessions between the UE and the cellular communication network infrastructure, the barring information message uniquely identifying at least one operator-defined group of network slices of the plurality of network slices by a slice group identifier mapped to a corresponding value field of the barring information message in accordance with a previously negotiated unified access control schema, the unified access control schema corresponding to an operator-defined access category that indicates a criteria type configured to support slice grouping; triggering, by the UE, establishment of a communication session between the UE and the cellular communication network infrastructure, the triggering invoking at least one network slice of the at least one operator-defined group of network slices; determining, by the UE based on the barring information message, whether the at least one network slice is barred from use in establishing communication sessions between the UE and the cellular communication network infrastructure; and establishing a communication session by the

UE with the cellular communication network infrastructure via a virtualized data network defined by the at least one network slice responsive only to the determining that the at least one network slice is not barred from use in establishing communication sessions between the UE and the cellular communication network infrastructure.

(10) According to another set of embodiments, a system is provided for unified access control of network slice groups in a cellular communication network infrastructure. The system includes: a plurality of virtualized data subnetworks built on the cellular communication network infrastructure, each virtualized data subnetwork uniquely being one of a plurality of network slices identifiable by a respective one of a plurality of network slice identifiers; and an access and mobility management function (AMF) node in communication with the plurality of virtualized data subnetworks and in communication with a plurality of user equipment devices (UEs) via one or more radio access networks (RANs). The AMF node includes: one or more processors; and non-transitory memory having instructions stored thereon, which, when executed, cause the one or more processors to perform steps. The steps include: negotiating a unified access control schema corresponding to an operator-defined access category that indicates a criteria type configured to support slice grouping; mapping a slice group identifier to a corresponding value field of a barring information message in accordance with the unified access control schema, the slice group identifier uniquely identifying an operator-defined group of network slices of the plurality of network slices; and communicating the barring information message to at least one of the plurality of UEs via the RAN to indicate which of the plurality of network slices is barred from use in establishing communication sessions between the at least one of the plurality of UEs and the cellular communication network infrastructure.

(11) This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used in isolation to determine the scope of the claimed subject matter. The subject matter should be understood by reference to appropriate portions of the entire specification of this patent, any or all drawings, and each claim.

(12) The foregoing, together with other features and embodiments, will become more apparent upon referring to the following specification, claims, and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is described in conjunction with the appended figures:

(2) FIG. 1 illustrates an embodiment of a cellular network system as context for various embodiments described herein;

(3) FIG. 2 shows a view of a conventional embodiment of a cellular communication network;

(4) FIGS. 3A and 3B show an example of a conventional schema for a network slice identifier;

(5) FIGS. 4A and 4B show an example of a novel schema for network slice identification that provides efficient slice grouping along the SST dimension and/or along the SD dimension, according to embodiments described herein;

(6) FIG. 5 shows an illustrative cellular communication network that supports wildcard-based slice grouping, according to various embodiments described herein;

(7) FIGS. 6A and 6B show illustrative flow diagrams for UE registration and subsequent establishment of one or more communication sessions on a cellular network that supports wildcard-enabled slice groups, according to various embodiments;

(8) FIG. 7 shows a flow diagram of a method to implement a first novel approach to unified access control over network slice groups, according to some embodiments described herein; and

(9) FIG. 8 shows a flow diagram of a method to implement a second novel approach to unified access control over network slice groups, according to some embodiments described herein.

(10) In the appended figures, similar components and/or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a second label (e.g., a lower-case letter) that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

DETAILED DESCRIPTION

(11) Embodiments of the disclosed technology will become clearer when reviewed in connection with the description of the figures herein below. In the following description, numerous specific details are set forth to provide a thorough understanding of the present invention. However, one having ordinary skill in the art should recognize that the invention may be practiced without these specific details. In some instances, circuits, structures, and techniques have not been shown in detail to avoid obscuring the present invention.

(12) Some modern wireless communication networks, such as fifth generation (5G) wireless networks, have begun to support so-called network slicing. The communication network includes physical (e.g., multi-domain) infrastructure, such as transport network infrastructure, core network infrastructure, and access network infrastructure. Virtual network operators (VNOs), also referred to as mobile network operators (MNOs), mobile virtual network operators (MVNOs), and the like can concurrently use the physical infrastructure to offer multiple virtualized networks. Such network slices can be implemented using various virtualization technologies, such as software-defined networking (SDN) and network functions virtualization (NFV) (also referred to as virtualized network functions (VNFs)). Each of those virtualized networks is considered to be a network slice, and each network slice can be designed (e.g., optimized) to support a particular desired set of users, applications, services, network functions, etc. Each network slice can be implemented with its own management domain and its own automation, and each network slice can be logically separated from other network slices. For example, different network slices can be used to separate and/or prioritize traffic, manage security, maintain separation between customers and/or services, etc.

(13) While the concept of network slicing was originally to enable self-contained virtualized networks, some VNOs have more recently found use cases for grouping network slices. For example, a VNO may offer multiple network slice types (e.g., optimized for particular categories of services) to each of multiple customers. When a particular user equipment (UE) device sends an attachment request to connect with the wireless network via an eNodeB, or other suitable network node, the VNO may attach the UE to the network concurrently via multiple network slices by associating the UE's communications with an explicit list of the identifier of each network slice that is part of the group. Particularly in cases where it is desirable to attach the UE to a large number of network slices, such as to all networks slices for a particular customer, such explicit listing of network slice identifiers can be inefficient.

(14) Embodiments described herein provide a novel slice identification schema that supports group-level identification of groups of network slices without individually listing slices in access-stratum and/or non-access stratum communications. In some implementations, network slices can be uniquely identified by a slice identifier that includes a slice/service type (SST) portion and a slice differentiator (SD) portion. As described herein, such a slice identifier can be configured to support one or more wildcard SST options and/or wildcard SD options to support wildcard-enabled slice groups. For example, setting the SST portion to the wildcard SST option can uniquely identify a slice group that includes all SSTs for a particular SD designated in the SD portion; and/or setting the SD portion to the wildcard SD option can uniquely identify a slice group that includes all SDs for a particular SST designated in the SST portion. Some embodiments enable use of such wildcard-enabled slice groups in context of unified access control. Some embodiments introduce a novel criteria type to unified access control to expressly support flexible slice grouping.

(15) For the sake of context, some embodiments are described herein with reference to various networking standards. In general, reference herein to a networking standard beginning with “TS” is intended to refer to a technical specification released by the 3rd Generation Partnership Project (3GPP) in connection with fifth generation (5G) cellular networks. However, such 5G-specific references are intended only to provide illustrative implementations for those contexts and are not intended to limit the application of embodiments described herein to other contexts. For example, innovative embodiments described herein can be applied beyond 5G networks, such as to any suitable previous and future generations and types of cellular and other wireless communication networks.

(16) FIG. 1 illustrates an embodiment of a cellular network system **100** (“system **100**”) as context for various embodiments described herein. System **100** can include a 5G New Radio (NR) cellular network; other types of cellular networks, such as 6G, **76**, etc. may also be possible. System **100** can include: user equipment (UE) **110** (UE **110-1**, UE **110-2**, UE **110-3**); base station **115**; cellular network **120**; radio units **125** (“RUs **125**”); distributed units **127** (“DUs **127**”); centralized unit **129** (“CU **129**”); 5G core **139**, and orchestrator **138**. FIG. 1 represents a component-level view. In an open radio access network (O-RAN), because components can be implemented as specialized software executed on general-purpose hardware, except for components that need to receive and transmit RF, the functionality of the various components can be shifted among different servers. For at least some components, the hardware may be maintained by a separate cloud-service provider, to accommodate where the functionality of such components is needed.

(17) UE **110** can represent various types of end-user devices, such as cellular phones, smartphones, cellular modems, cellular-enabled computerized devices, sensor devices, gaming devices, access points (APs), any computerized device capable of communicating via a cellular network, etc. Generally, UE can represent any type of device that has an incorporated 5G interface, such as a 5G modem. Examples can include sensor devices, Internet of Things (IoT) devices, manufacturing robots; unmanned aerial (or land-based) vehicles, network-connected vehicles, etc. Depending on the location of individual UEs, UE **110** may use RF to communicate with various base stations of cellular network **120**. As illustrated, two base stations are illustrated: base station **121** can include: structure **115-1**, RU **125-1**, and DU **127-1**. Structure **115-1** may be any structure to which one or more antennas (not illustrated) of the base station are mounted. Structure **115-1** may be a dedicated cellular tower, a building, a water tower, or any other man-made or natural structure to which one or more antennas can reasonably be mounted to provide cellular coverage to a geographic area. Similarly, base station **121-2** can include: structure **115-2**, RU **125-2**, and DU **127-2**.

(18) Real-world implementations of system **100** can include many (e.g., thousands) of base stations and many CUs and 5G core **139**. BS **115** can include one or more antennas that allow RUs **125** to communicate wirelessly with UEs **110**. RUs **125** can represent an edge of cellular network **120** where data is transitioned to wireless communication. The radio access technology (RAT) used by RU **125** may be 5G New Radio (NR), or some other RAT. The remainder of cellular network **120** may be based on an exclusive 5G architecture, a hybrid 4G/5G architecture, a 4G architecture, or some other cellular network architecture. Base station equipment **121** may include an RU (e.g., RU **125-1**) and a DU (e.g., DU **127-1**).

(19) One or more RUs, such as RU **125-1**, may communicate with DU **127-1**. As an example, at a possible cell site, three RUs may be present, each connected with the same DU. Different RUs may be present for different portions of the spectrum. For instance, a first RU may operate on the spectrum in the citizens broadcast radio service (CBRS) band while a second RU may operate on a separate portion of the spectrum, such as, for example, band **71**. One or more DUs, such as DU **127-1**, may communicate with CU **129**. Collectively, an RU, DU, and CU create a gNodeB, which serves as the radio access network (RAN) of cellular network **120**. CU **129** can communicate with 5G core **139**. The specific architecture of cellular network **120** can vary by embodiment. Edge cloud server systems outside of cellular network **120** may communicate, either directly, via the

Internet, or via some other network, with components of cellular network **120**. For example, DU **127-1** may be able to communicate with an edge cloud server system without routing data through CU **129** or 5G core **139**. Other DUs may or may not have this capability.

(20) While FIG. **1** illustrates various components of cellular network **120**, other embodiments of cellular network **120** can vary the arrangement, communication paths, and specific components of cellular network **120**. While RU **125** may include specialized radio access componentry to enable wireless communication with UE **110**, other components of cellular network **120** may be implemented using either specialized hardware, specialized firmware, and/or specialized software executed on a general-purpose server system. In an O-RAN arrangement, specialized software on general-purpose hardware may be used to perform the functions of components such as DU **127**, CU **129**, and 5G core **139**. Functionality of such components can be co-located or located at disparate physical server systems. For example, certain components of 5G core **139** may be co-located with components of CU **129**.

(21) In a possible O-RAN implementation, DUs **127**, CU **129**, 5G core **139**, and/or orchestrator **138** can be implemented virtually as software being executed by general-purpose computing equipment, such as in a data center. Therefore, depending on needs, the functionality of a DU, CU, and/or 5G core may be implemented locally to each other and/or specific functions of any given component can be performed by physically separated server systems (e.g., at different server farms). For example, some functions of a CU may be located at a same server facility as where the DU is executed, while other functions are executed at a separate server system. In the illustrated embodiment of system **100**, cloud-based cellular network components **128** include CU **129**, 5G core **139**, and orchestrator **138**. Such cloud-based cellular network components **128** may be executed as specialized software executed by underlying general-purpose computer servers. Cloud-based cellular network components **128** may be executed on a third-party cloud-based computing platform or a cloud-based computing platform operated by the same entity that operates the RAN. A cloud-based computing platform may have the ability to devote additional hardware resources to cloud-based cellular network components **128** or implement additional instances of such components when requested.

(22) Components such as DUs **127**, CU **129**, orchestrator **138**, and 5G core **139** may include various software components that are required to communicate with each other, handle large volumes of data traffic, and are able to properly respond to changes in the network. In order to ensure not only the functionality and interoperability of such components, but also the ability to respond to changing network conditions and the ability to meet or perform above vendor specifications, significant testing must be performed.

(23) Kubernetes, or some other container orchestration platform, can be used to create and destroy the logical CU or 5G core units and subunits as needed for the cellular network **120** to function properly. Kubernetes allows for container deployment, scaling, and management. As an example, if cellular traffic increases substantially in a region, an additional logical CU or components of a CU may be deployed in a data center near where the traffic is occurring without any new hardware being deployed. (Rather, processing and storage capabilities of the data center would be devoted to the needed functions.) When the need for the logical CU or subcomponents of the CU no longer exists, Kubernetes can allow for removal of the logical CU. Kubernetes can also be used to control the flow of data (e.g., messages) and inject a flow of data to various components. This arrangement can allow for the modification of nominal behavior of various layers.

(24) The deployment, scaling, and management of such virtualized components can be managed by orchestrator **138**. Orchestrator **138** can represent various software processes executed by underlying computer hardware. Orchestrator **138** can monitor cellular network **120** and determine the amount and location at which cellular network functions should be deployed to meet or attempt to meet service level agreements (SLAs) across slices of the cellular network.

(25) A network slice functions as a virtual network operating on cellular network **120**. Cellular

network **120** is shared with some number of other network slices, such as hundreds or thousands of network slices. Communication bandwidth and computing resources of the underlying physical network can be reserved for individual network slices, thus allowing the individual network slices to reliably meet defined SLA parameters. By controlling the location and amount of computing and communication resources allocated to a network slice, the QoS and QoE for UE can be varied on different slices. A network slice can be configured to provide sufficient resources for a particular application to be properly executed and delivered (e.g., gaming services, video services, voice services, location services, sensor reporting services, data services, etc.). However, resources are not infinite, so allocation of an excess of resources to a particular UE group and/or application may be desired to be avoided. Further, a cost may be attached to cellular slices: the greater the amount of resources dedicated, the greater the cost to the user; thus optimization between performance and cost is desirable.

(26) Particular network slices may only be reserved in particular geographic regions. For instance, a first set of network slices may be present at RU **125-1** and DU **127-1**, a second set of network slices, which may only partially overlap or may be wholly different from the first set, may be reserved at RU **125-2** and DU **127-2**.

(27) Further, particular cellular network slices may include some number of defined layers. Each layer within a network slice may be used to define QoS parameters and other network configurations for particular types of data. For instance, high-priority data sent by a UE may be mapped to a layer having relatively higher QoS parameters and network configurations than lower-priority data sent by the UE that is mapped to a second layer having relatively less stringent QoS parameters and different network configurations.

(28) FIG. 2 shows a view of a conventional embodiment of a cellular communication network **200**. For the sake of simplicity, the conventional network **200** is shown with a single UE **110** in communication with the network infrastructure via a single tower **115**. The tower **115** can generally represent a radio access network (RAN), such as an eNodeB, or the like. The network infrastructure is illustrated generally as a physical infrastructure **210** having subnetworks associated with virtual network operators (VNOs) **220**. As described herein, each VNO **220** can implement multiple (e.g., a large number of) network slices **230**, each supporting a stand-alone virtual network with its own network management, orchestration, etc. that is optimized or otherwise configured for supporting particular applications, providing particular services, etc.

(29) UEs **110** can receive tailored communication services via the RAN and the network slices **230**, but the UEs **110** and the RAN are not considered part of any network slices **230**. Further, depending on the physical and logical architecture, certain network functions of the network infrastructure can be common to the network slices **230**, while other network functions can be dedicated to particular network slices **230**. As one example, the network infrastructure can include a number of common network functions, such as a network slice selection function (NSSF) and a network repository function (NRF). The example network infrastructure also includes a number of slice-specific functions, such as each network slice **230** having its own instance of a session management function (SMF) to provide the slice with management functions, and its own instance of a user plane function (UPF) to provide a respective tailored data network via the network slice **230**. Some network functions can be common from the UE **110** side, but slice-specific from the network side, or vice versa. For example, the example network infrastructure may dedicate different instances of an access and mobility management function (AMF) to different network slices **230**, but the AMF may be a common node from the perspective of any particular UE **110**.

(30) According to some network standards, the behavior of a network slice **230** is realized by a network slice instance (NSI), which is an activated network slice. Each NSI can include one or more network slice subnet instances (NSSIs), such as one or more core network NSSIs and one or more access network NSSIs. Each network slice can be uniquely identified. In the illustrated conventional network **200**, each network slice is uniquely identified by a single-network slice

selection assistance information (S-NSSAI) identifier **235**. The S-NSSAI **235** includes two portions: a slice/service type (SST) portion **237** and a slice differentiator (SD) portion **239**. Example schemas for defining network slice identifiers in context of 5G networks are discussed in TS 23.501 as promulgated by 3GPP.

(31) Each VNO **220** can offer a collection of network services defined by a public land mobile network (PLMN). According to the PLMN, each UE **110** can be provisioned with a “configured NSSAI,” which can correspond to a collection of S-NSSAIs **235**. When a UE **110** desires to establish a communication session over the illustrated conventional network **200**, the UE **110** can derive a “requested” NSSAI from its configured NSSAI. The UE **110** can send a UE registration request message **202** to the RAN (e.g., to cell tower **115**), identifying the requested NSSAI. The RAN can be aware of which AMFs support which network slices **230**, and the RAN can select an appropriate AMF to support the requested NSSAI, accordingly. The AMF can derive an “allowed” NSSAI (e.g., and a “rejected” NSSAI) on its own, or in conjunction with the NSSF. Information about the allowed and rejected NSSAIs is sent back to the UE **110**. The UE **110** can then request and establish one or more PDU sessions over one or more of the S-NSSAIs **235** of the allowed NSSAI.

(32) As described herein, each network slice **230** can be configured by the VNO **220** to be tailored to a particular type of service, application, etc. For example, when the UE **110** requests to establish a PDU session for a particular application, the request can indicate a particular S-NSSAI **235** (of the allowed NSSAI) that is most suitable for, tailored for, or otherwise appropriate for that application. For example, each VNO **220** can create and manage a large number of network slices **230**, each tailored for a particular service type (indicated by the SST portion **237** of the S-NSSAI **235**) and for a particular tenant or type of tenant (indicated by the SD portion **239** of the S-NSSAI **235**). The tenant or type of tenant can be a category of subscribers, an enterprise customer, etc.

(33) For added context, FIGS. 3A and 3B show an example of a conventional schema **300** for a S-NSSAI **235**. Such a schema is defined in various wireless communication networking standards. As described above, the S-NSSAI **235** includes an SST portion **237** and an SD portion **239**. Each SST portion **237** defines a category of service for which the network slice is being configured or optimized as corresponding to expected performance and behavior of the virtual network provided by the network slice **230**. Each SD portion **239** provides optional differentiation between network slices **230** of the same SST, such as to allow segmentation between network slices **230** of the same SST being provided to different customers. According to some standards, the SST portion **237** is represented by one octet (an SST value **334**), such that there can be up to 256 SST values **334** that can each have an associated SST name **332** and SST service description **336**; and the SD portion **239** is represented by three octets (an SD value **344**), such that there can be over 10 million SD values **334** that can each have an associated SD description **346**. The range of values is indicated as between hexadecimal ‘#000000’ (i.e., ‘0’ decimal) and ‘#FFFFFF’ (i.e., ‘16,777,215’ decimal).

(34) As shown in FIG. 3A, according to some wireless communication standards, there is a first set of SST values **334** (‘00000001’ through ‘00000101’ binary, or ‘1’ through ‘5’ decimal) that have defined SST names **332** and SST service descriptions **336**. SST value **334** ‘1’ corresponds to enhanced (or extreme) mobile broadband (eMBB) network slices **230**, which can be optimized for broadband video applications, and/or other types of applications that tend to consume high amounts of network bandwidth. SST value **334** ‘2’ corresponds to ultra-reliable low-latency communications (URLLC) network slices **230**, which can be optimized for certain emergency applications, telemedicine applications, and/or other contexts that tend to require very high reliability and very low latency. SST value **334** ‘3’ corresponds to massive Internet-of-things (MIoT) network slices **230** (also referred to as massive machine-type communications, MMTC), which can be optimized for IoT sensor devices and applications, and/or other context in which relatively large numbers of devices each tend to use relatively small amounts of network bandwidth and/or do not require high performance. SST value **334** ‘4’ corresponds to vehicle-to-X (V2X) network slices **230**, which can

be optimized for applications and services to support autonomous vehicles, and/or the like. SST value **334** '5' corresponds to high-performance machine-type communications (HMTC) network slices **230**, which can be optimized for IoT and/or other devices that tend to require high throughput and high reliability. A second set of SST values **334** ('00000110' through '01111111' binary, or '6' through '127' decimal) is reserved for future definition by the standard. A third set of SST values **334** ('10000000' through '11111111' binary, or '128' through '255' decimal) is allocated for definition by the VNO **220**.

(35) As shown in FIG. 3B, according to some wireless communication standards, all of the SD values **344** can be defined by the VNO **220**. Each SD value **344** can be associated by the VNO **220** with a corresponding SD description **346**. For example, the SD description **346** can indicate a particular tenant (e.g., enterprise subscriber, etc.), class of service, or any other suitable designation that differentiates the particular network slice **230** from another network slice **230** of the same SST. Suppose, for example, that a large VNO **220** has a number of subscribers, including a large vehicle company (Company A), a large retailer (Company B), and a large logistics company (Company C). The VNO **220** may decide to build out network services for each of these subscribers to include eMBB network slices **230** for all three companies, V2X network slices **230** for Company A and Company C, MIoT network slices **230** for Company B and Company C, etc. In such a case, the VNO **220** now has eMBB network slices **230** for three different customers, and the VNO **220** can use different SD values **344** to provide segmentation and differentiation between those network slices **230**.

(36) In some cases, it may be desirable to treat network slices **230** as a group with respect to either a common SST dimension or a common SD. As one illustration, a VNO **220** may desire to perform a slice orchestration or management function on a particular SST, such as by upgrading certain network functions for all subscribers that have been allocated network slices **230** of that particular SST. For example, the VNO **220** has multiple car companies as subscribers, and each car company is allocated at least one V2X network slice **230**, differentiated by respective SD values **344**. Performing a function to affect all network slices **230** of the same SST may involve repeating related messaging by the VNO **220** among components of the network for each of the effected network slices **230**. For example, a particular message operates on a designated network slice **230** by indicating the corresponding S-NSSAI **235** of the network slice **230**. If there are twenty network slices **230** of the same SST, each differentiated by a respective SD value **344**, it may be necessary to repeat the message for each of the twenty differentiated network slices **230**. As another illustration, a VNO **220** may desire to perform a slice orchestration or management function for a particular subscriber, such as by upgrading certain network functions for all network slices **230** associated with that subscriber's SD value **344**. As in the previous illustration, performing a function to affect all network slices **230** of the same SD may involve repeating related messaging by the VNO **220** among components of the network for each of the effected network slices **230**. For example, a particular message operates on a designated network slice **230** by indicating the corresponding S-NSSAI **235** of the network slice **230**. If there are twenty network slices **230** of the same SD, each of a different respective SST value **334**, it may be necessary to repeat the message for each of the twenty network slices **230**.

(37) Such repeated messaging can be inefficient and/or otherwise undesirable. FIGS. 4A and 4B show an example of a novel schema **400** for network slice identification that provides efficient slice grouping along the SST dimension and/or along the SD dimension, according to embodiments described herein. For ease of comparison, the novel slice identification schema **400** is shown in a similar manner to the conventional schema **300** of FIGS. 3A and 3B. In particular, the novel slice identification schema **400** is illustrated as backwards-compatible with the conventional schema **300** of the S-NSSAI **235**.

(38) FIG. 4A shows an illustrative listing of SST information, including SST values **334** with corresponding SST names **332** and SST service descriptions **336**. Unlike in the conventional

schema, the novel slice identification schema **400** includes at least one wildcard SST option **410**. The wildcard SST option **410** is assigned to a particular SST value **334**, illustrated as 'XXX' (with a corresponding SST name **332** and SST service description **336**). In some implementations, the wildcard SST option **410** is assigned to SST value **334** '0' (i.e., '00000000' binary), which is unassigned in applicable standards as of this filing. In other implementations, the wildcard SST option **410** is assigned to any of the set of SST values **334** from '6' to '127', which are reserved for designation in the standard, but are as-yet unassigned (i.e., in the range of SST values **334** from '00000110' to '01111111' binary). Whether the wildcard SST option **410** is assigned to SST value **334** '0', or to any of the unassigned SST values **334** between '6' and '127', such implementations involve inclusion of the assignment in the relevant wireless communication standards. In other implementations, the wildcard SST option **410** is assigned to any of the set of SST values **334** from '128' to '255', which are reserved for designation by VNOs **220** without requiring acceptance by any standards organization (i.e., in the range of SST values **334** from '10000000' to '11111111' binary).

(39) FIG. 4B shows an illustrative listing of SD information, including SD values **344** with corresponding SD descriptions **346**. Unlike in the conventional schema, the novel slice identification schema **400** includes at least one wildcard SD option **420**. The wildcard SD option **420** is assigned to a particular SD value **344**, illustrated as '#XXXXXX' (with a corresponding SD description **346**). As described above, current standards allocate all of the more than 10 million possible SD values **344** for designation by the VNOs **220** without requiring acceptance by any standards organization. Implementations can assign the wildcard SD option **420** to any of over 16 million SD values **344**.

(40) In some implementations, the wildcard SST option **410** indicates "all SSTs." When the S-NSSAI **235** has the SST portion **237** set to the wildcard SST option **410** and the SD portion **239** set to a particular SD value **344**, the S-NSSAI **235** directly refers to a slice group that includes the network slices **230** for all SSTs for that particular SD. Similarly, in some implementations, the wildcard SD option **420** indicates "all SDs." When the S-NSSAI **235** has the SD portion **239** set to the wildcard SD option **420** and the SST portion **237** set to a particular SST value **334**, the S-NSSAI **235** directly refers to a slice group that includes the network slices **230** for all SDs for that particular SST. Any type of network slice **230** grouping that is enabled by the described types of wildcard SST options **410** and/or wildcard SD options **420** can be referred to as a wildcard-enabled slice group.

(41) While each of FIGS. 4A and 4B shows only a single respective wildcard option, other implementations can include other numbers of wildcard SST options **410** and/or wildcard SD options **420**. For example, as noted above, some implementations use the set of SST values **334** reserved for VNO **220** designation (i.e., SST values **334** from '10000000' to '11111111' binary) for the wildcard SST option **410**. In such an implementation, one or more subsets of those SST values **334** can be related for grouping purposes, and one of the SST values **334** in each subset can be used as the wildcard SST option **410** for that subset. For example, the VNO **220** can create a set of 15 network slices **230**, all tailored for different types of network security approaches in the range of SST values **334** from '10010001' to '10011111' (binary), and SST value **334** '10010000' can be assigned as the wildcard SST option **410** for that subset; setting the SST value **334** to '10010000' directly indicates a slice group of the 15 network slices **230**. Such an approach can be used to assign, for example, a first wildcard SST option **410** to group all SSTs (e.g., for a particular SD) and one or more additional wildcard SST options **410** to group subsets of SSTs. The same technique can be used to assign (e.g., in addition to, or instead of a wildcard SD option **410** to group all SDs) one or more wildcard SD options **420** to group subsets of SDs.

(42) FIG. 5 shows an illustrative cellular communication network **500** that supports wildcard-based slice grouping, according to various embodiments described herein. For the sake of simplicity, the network **500** is shown with a single UE **110** in communication with the network infrastructure via a

single tower **115**. The tower **115** can generally represent a radio access network (RAN), such as an eNodeB, or the like. The network infrastructure is illustrated generally as a physical infrastructure **210** having subnetworks associated with virtual network operators (VNOs) **220**. Only a single VNO **220** is shown, but the physical network infrastructure **210** can support any suitable number of VNOs **220**. Each VNO **220** can implement multiple (e.g., a large number of) network slices **230**, each supporting a stand-alone virtual network with its own network management, orchestration, etc. that is optimized or otherwise configured for supporting particular applications, providing particular services, etc. Additionally, as shown, wildcard SST options **410** and/or wildcard SD options **420** can be used to implement slice grouping for the network slices **230** of a VNO **220**. (43) For example, each network slice **230** is uniquely identified by a unique slice identifier. The slice identifier can be implemented using any suitable schema that supports the types of wildcard SST options **410** and/or wildcard SD options **420** described herein. Some embodiments implement the unique slice identifiers as single-network slice selection assistance information (S-NSSAI) identifiers **235**, each having an SST portion **237** and an SD portion **239**. By using the types of wildcard SST options **410** and/or wildcard SD options **420** described herein, such communications can directly invoke a correspondingly defined group of network slices **230** (i.e., a wildcard-enabled slice group). As used herein, a network slice **230** is considered to be “invoked” by a communication when the communication includes an identification of the network slice **230** either directly, as part of a NSSAI, as part of a wildcard-enabled slice group, etc.

(44) Several examples of such wildcard-enabled slice groups are shown. In the illustrated examples, the VNO **220** has created a large number of slices to support up to K SSTs for up to N SDs. For example, network slice **230-11** is tailored for the 1st SST for the 1st SD, and network slice **230-NK** is tailored for the Kth SST for the Nth SD. The illustrated arrangement of network slices **230** is simplified in a number of ways to avoid over-complicating the figure. For example, the VNO **220** can typically provide network slices **230** of different sets of SSTs for different SDs. One illustrated example of wildcard-enabled slice groups are groupings in the SST dimension. For example, the “SST1 Slice Group” **520-1** indicates a grouping of all network slices **230** common to a first SST, regardless of the SD (i.e., for all SDs having such a network slice **230**). In the S-NSSAI **235** for the SST1 Slice Group **520-1**, the SST portion **237** indicates an SST value **334** of ‘1’, and the SD portion **239** indicates an SD value **344** corresponding to the (e.g., or a) wildcard SD option **410**. Another illustrated example of wildcard-enabled slice groups are groupings in the SD dimension. For example, the “SD1 Slice Group” **510-1** indicates a grouping of all network slices **230** common to a first SD, regardless of the SST. In the S-NSSAI **235** for the SD1 Slice Group **510-1**, the SST portion **237** indicates an SST value **334** corresponding to the (e.g., or a) wildcard SST option **410**, and the SD portion **239** indicates an SD value **344** of ‘1’.

(45) At least some of the communications between UEs **110**, RAN components (e.g., tower **115**), physical network infrastructure **210**, virtual network functions, etc. include explicit slice identification. For example, communications regarding registration of UEs and establishment of a communication session, communications regarding slice management, and other types of communications include identification of S-NSSAIs **235** to indicate which network slice **230** or network slices **230** are invoked by the communications. For the sake of illustration, the network **500** shows a representation of UE registration and response messaging. The UE **110** communicates a UE registration request message **202** to the RAN (e.g., to cell tower **115**), which can pass along the message (e.g., with modification) to the network functions of the VNO **220**. In response, a UE registration response message **504** is communicated to the UE **110** via the RAN to indicate at least one allowed NSSAI representing a set of S-NSSAIs **235**. By optionally exploiting one or more wildcard SST options **410** and/or wildcard SD options **420**, the set of S-NSSAIs **235** of the at least one allowed NSSAI can include S-NSSAIs **235** corresponding to individual network slices **230** (e.g., **504-1**), S-NSSAIs **235** corresponding to wildcard-enabled slice groups that are SD slice groups **510** (e.g., **504-2**), and/or S-NSSAIs **235** corresponding to wildcard-enabled slice groups that

are SST slice groups **520** (e.g., **504-3**).

(46) FIGS. **6A** and **6B** show illustrative flow diagrams **600** for UE registration and subsequent establishment of one or more communication sessions on a cellular network that supports wildcard-enabled slice groups, according to various embodiments. Turning first to FIG. **6A**, a portion of the cellular network is shown to include a UE **110**, a RAN **610**, and common network functions (at least from the perspective of the UE **110**). The common network functions can include at least one or more access and mobility management functions (AMFs) **620**, a unified data management (UDM) function **630**, and a network slice selection function (NSSF) **640**.

(47) Embodiments begin at stage **604** by the UE **110** deriving a “requested” NSSAI from its configured NSSAI. For example, each VNO **220** can offer a collection of network services defined by a public land mobile network (PLMN). According to the PLMN, each UE **110** can be provisioned with a “configured NSSAI,” which can correspond to a collection of S-NSSAIs **235**. The UE **110** can then derive its requested NSSAI from the configuration of the PLMN. The UE **110** can then send a UE registration request message **202-1** to the RAN **610**. At stage **608**, the RAN **610** can select an appropriate AMF **620**. For example, the RAN **610** is aware of which network slices **230** are supported by each AMF **620** based on information exchanged during an “N2/NG” setup procedure, or the like. Thus, the RAN **610** can select an appropriate AMF **620** based on the set of S-NSSAIs that correspond to the requested NSSAI and based on the RAN's **610** knowledge of which of those S-NSSAIs are supported by which AMF **620**.

(48) At stage **612**, the AMF **620** can determine one or more “allowed” NSSAIs and/or one or more “rejected” NSSAIs. In some cases, the AMF **620** can make this determination on its own. In other cases, the AMF **620** makes this determination based on interrogating the NSSF **640**. In some cases, the AMF **620** communicates and received subscription messages **605** with the UDM **630** to determine to which NSSAIs the UE **110** is subscribed. As described above, each allowed (and/or rejected) NSSAI includes a set of corresponding S-NSSAIs **235**, and one or more of the S-NSSAIs **235** can identify a wildcard-enabled slice group. The AMF can communicate a UE registration response message **504** back to the UE **110** via the RAN **610**. The UE registration response message **504** indicates the allowed and/or rejected NSSAIs. Based on the UE registration response message **504**, at stage **616**, the UE **110** can request establishment of a communication session via one or more network slices **230**. For example, the UE **110** can request a so-called “PDU session,” to provide end-to-end connectivity between the UE **110** and the specific data network or group of data networks that is implemented via the user plane function (UPF) or group of UPFs corresponding to a S-NSSAI **235** (e.g., representing a single network slice **230** or a wildcard-enabled slice group).

(49) Turning to FIG. **6B**, a different portion of the cellular network is shown to include the UE **110**, the RAN **610**, and the AMF **620** of FIG. **6A**, as well as one or more UPFs **650** and one or more session management functions (SMF) **660**. As noted above, at stage **616**, the UE **110** can request establishment of a communication session via one or more network slices **230**. Such a request can involve sending a session establishment request message **625** that indicates a specific S-NSSAI **235** over which to establish the session. For example, the UE **110**, or a user of the UE **110**, can start an application (e.g., a smartphone app) that can be mapped to a specific S-NSSAI **235** based on a UE route selection policy (URSP), or the like. The session establishment request message **625** can be communicated from the UE **110** via the RAN **610** to the AMF **620** previously determined by the RAN **610** to support the allowed NSSAI (that includes the S-NSSAI **235** being requested).

(50) As noted above, a particular network slice **230** can effectively provide a stand-alone virtualized data network implemented generally by at least a UPF **650** and a SMF **660**. At stage **624**, the AMF **620** can select one or more SMFs **660** to support session management for the requested session. In cases where the S-NSSAI **235** indicates a wildcard-enabled slice group, more than one SMF **660** may be invoked. In some cases, the AMF **620** can directly select the appropriate one or more SMFs **660**. If the AMF **620** is unable to autonomously select an appropriate one or more SMFs **660** (e.g., the AMF **620** is not aware of SMF information in the network), the AMF **620**

can engage a network function (NF) discovery routine with a network repository function (NRF) and/or engage a slice selecting service of the NSSF **640**.

(51) Having selected one or more appropriate SMFs **660**, the AMF **620** can communicate session creation request and response messages **635** with the selected SMF(s) **660**. At stage **628**, the SMF(s) **660** can select one or more appropriate UPFs **650** with which to establish the requested session based on the requested S-NSSAI **235**. The SMF(s) **660** and UPF(s) **650** can exchange session establishment messages **645** by which to establish the requested session for the UE **110**. According to some network configurations, the UPF(s) **650** are not directly aware of the network slices **230**, such that the session establishment messages **645** do not explicitly indicate S-NSSAIs **235**. Instead, the SMF(s) **660** signals one or more network instances to the UPF(s) **650** via the session establishment messages **645**, and the UPF(s) **650** distinguish the domains of the network slices **230** based on the communicated network instances.

(52) While certain descriptions above focus on certain types of messaging that explicitly identify an S-NNSAI **235**, other embodiments relate to adaptations of so-called “unified access control” to allow for slice grouping. In general, unified access control allows the VNOs **220** to control network access, such as to determine which access attempts to allow or block in response to network congestion. Typically, such unified access control is based on so-called “access identities” and “access categories.” VNOs **220** can broadcast barring information that identifies lists of barring parameters associated with access identities and access categories. When a UE **110** desires to access the network, the access attempt can have a corresponding one or more access identities and an access category, and the UE **110** can determine whether it is allowed to make the access attempt based on comparing the corresponding access identity and access category to the previously broadcast barring information. As such, the unified access control allows a VNO **220** to permit or bar network access by any UE **110**. Some access identities and access categories can be defined by applicable network standards, and other can be reserved for definition and assignment by the VNOs **220**. VNO-defined unified access control parameters (e.g., VNO-defined access categories) can be specified in the serving PLMN.

(53) FIGS. **7** and **8** shows illustrative flow diagrams for methods **700** and **800**, respectively, of performing unified access control with slice groups, according to various embodiments. The methods **700** and **800** can be implemented in any suitable communication network, such as those described above. For the sake of illustration, it is assumed that the methods **700** and **800** involve non-access stratum (NAS) communications in a cellular network, such as between a UE **110** and an AMF **620**. In such contexts, the AMF **620** can communicate unified access control messaging **705** to the UE **110**, such as by broadcasting barring information that indicates access identities and/or access categories that are barred from access to the network. Conventional network standards presently include an access category to permit unified access control at the network slice **230** level, but there is no support for unified access control at a slice group level. For example, Table 9.11.3.38.1 of TS 23.501, as promulgated by 3GPP, describes a number of criteria types that can be used in operator-defined access categories.

(54) Among the standard-defined criteria types is a “S-NSSAI Type.” This criteria type is defined to allows the VNO **220** to use unified access control to bar an individual S-NSSAI **235**, or a list of individual S-NSSAIs **235**. For example, each time the AMF **620** communicates barring information to bar access to multiple network slices, the AMF **620** the barring information includes a list of the individual S-NSSAIs **235** for those network slices. Such an approach can permit the VNO **220** to flexibly define any list of network slices. However, including a list of all barred and non-barred S-NSSAIs in all relevant NAS messaging can also be inefficient.

(55) Embodiments described herein provide two novel approaches to unified access control of network slice groups in a cellular communication network infrastructure. Both approaches generally begin by negotiating (e.g., between an access and mobility management function (AMF) and a user equipment device (UE) via a radio access network (RAN)) a unified access control

schema corresponding to an operator-defined access category that indicates a criteria type configured to support slice grouping. The approaches can then map a slice group identifier to a corresponding value field of a barring information message in accordance with the unified access control schema, the slice group identifier uniquely identifying an operator-defined group of network slices of a plurality of network slices, each of the plurality of network slices being uniquely identifiable by a network slice identifier separate from the slice group identifier. The approaches can then communicate the barring information message from the AMF to the UE via the RAN to indicate to the UE which of the plurality of network slices is barred from use in establishing communication sessions between the UE and the cellular communication network infrastructure.

(56) According to both approaches, the UE ultimately receives the barring information message (e.g., and locally stores barring information encoded therein) indicating to the UE which of a plurality of network slices is barred from use in establishing communication sessions between the UE and the cellular communication network infrastructure. At some subsequent time, the UE trigger establishment of a communication session between the UE and the cellular communication network infrastructure, and the triggering can invoke at least one network slice of the at least one operator-defined group of network slices. The UE can determine, based on the barring information message, whether the at least one network slice is barred from use in establishing communication sessions between the UE and the cellular communication network infrastructure. The UE can then establish a communication session with the cellular communication network infrastructure (via a virtualized data network defined by the at least one network slice responsive) only after determining that the invoked network slice is not barred in the barring information.

(57) FIG. 7 shows a flow diagram of a method **700** to implement a first novel approach to unified access control over network slice groups, according to some embodiments described herein. Such an approach uses an S-NSSAI criteria type as part of a VNO-defined access category with its contents (e.g., value fields) configured to support wildcard-enabled slice groups. Embodiments of the method **700** can begin at stage **704** by negotiating a unified access control schema of the S-NSSAI type. Such a negotiation can be performed, for example, between the AMF **620** and one or more UEs **110** during UE **110** registration with the network, or at any other suitable time. The negotiation can define a schema by which to indicate those of the network slices that are barred and those that are allowed. In general, the schema can define value fields that can be evaluated (e.g., interpreted) by a receiving UE **110** once the UE **110** is made aware of the schema. For example, the schema can define the manner in which S-NSSAIs **235** are indicated (e.g., in which octet, in which bit positions, in what format, in which order, how many bits per S-NSSAI, etc.) and the manner of indicating whether a particular S-NSSAI is barred (e.g., a bit value of '1' can indicate that a particular S-NSSAI **235** mapped to that bit position is barred from access by an associated UE **110**).

(58) At stage **708**, embodiments can map each of multiple S-NSSAIs to corresponding value fields of barring information in accordance with the schema, where at least one of the mapped S-NSSAIs uniquely identifies a wildcard-enabled slice group. For example, the AMF **620** can inform the UE **110** of a mapping between a number of S-NSSAIs **235** and some number of bits. Subsequently, the AMF **620** can broadcast barring information by setting some of the bits to '1' and other bits to '0' (e.g., '1' indicates that UE **110** is barred from accessing the S-NSSAI **235** mapped to that bit, and '0' indicates that the UE **110** is allowed access to the S-NSSAI **235** mapped to that bit; or vice versa). In this way, any device that is not registered with the AMF **620** may be able to pick up the broadcast barring information, but will be unable to interpret the information without the schema received during the registration.

(59) In the mapping of stage **708**, one or more of the S-NSSAIs **235** that is mapped to bits (or otherwise to value fields) by the AMF **620** includes one or more wildcard SST options **410** and/or wildcard SD options **420** to indicate one or more SST slice groups **520** and/or SD slice groups **510**.

At stage **712**, embodiments can communicate the barring information to the UE **110** to indicate whether the UE **110** is barred from establishing a communication session via the indicated wildcard-enabled slice group. For example, the AMF **620** broadcasts the barring information to a UE **110** including setting the one or more bits corresponding to the wildcard-enabled slice group to a barring state (e.g., to a '1' or '0' to indicate that the corresponding S-NSSAI **235** is barred). Doing so bars the entire wildcard-enabled slice group (i.e., the SST slice group **520** or SD slice group **510**) uniquely defined by the S-NSSAI **235** without explicitly listing all network slices that are members of the group.

(60) At stage **720**, embodiments of the UE **110** can receive and locally store the barring information. At some subsequent time, at stage **724**, the UE **110** can trigger a session establishment routine for establishing a PDU session, or other suitable communication session, between the UE **110** and the network using the network slice identified by a particular S-NSSAI **235**. As part of the routine, at stage **728**, the UE **110** can determine whether the requested S-NSSAI **235** is barred according to the barring information received and stored at stage **720**. If the S-NSSAI **235** is not barred, a session can be established via the requested S-NSSAI **235** (stage **732**). If the S-NSSAI **235** is barred, a session cannot be established via the requested S-NSSAI **235** (stage **734**).

(61) FIG. **8** shows a flow diagram of a method **800** to implement a second novel approach to unified access control over network slice groups, according to some embodiments described herein. Such an approach uses a novel "slice group criteria type" as part of a VNO-defined access category with its contents (e.g., value fields) configured to support VNO-defined slice groups. For example, according to Table 9.11.3.38.1 of TS 23.501, the S-NSSAI criteria type is invoked by setting the eighth octet of the VNO-defined access category definition to '00000010'. The criteria type octet can be followed by one or more S-NSSAI length-value value fields. For example, each value field can include one octet to indicate the length (in octets) of a following S-NSSAI value field, followed by the number of octets of the S-NSSAI value field to identify the S-NSSAI (e.g., one octet for the SST value and three octets for the SD value, as described above).

(62) Embodiments of the method **800** can begin at stage **804** by negotiating a unified access control schema of the slice group criteria type. Such a negotiation can be performed, for example, between the AMF **620** and one or more UEs **110** during UE **110** registration with the network, or at any other suitable time. The negotiation can define a schema by which to define VNO-defined slice groups (i.e., at least to identify which network slices are part of any particular VNO-defined slice group) and by which to indicate which of the VNO-defined slice groups are barred or allowed. In general, the schema can define value fields that can be evaluated (e.g., interpreted) by a receiving UE **110** once the UE **110** is made aware of the schema.

(63) At stage **808**, embodiments can map each of multiple VNO-defined slice groups to corresponding value fields of barring information in accordance with the schema. For example, the AMF **620** can inform the UE **110** of a mapping between a number of VNO-defined slice group and some number of bits. Subsequently, the AMF **620** can broadcast barring information by setting some of the bits to '1' and other bits to '0' (e.g., '1' indicates that UE **110** is barred from accessing the S-NSSAI **235** mapped to that bit, and '0' indicates that the UE **110** is allowed access to the S-NSSAI **235** mapped to that bit; or vice versa). In this way, any device that is not registered with the AMF **620** may be able to pick up the broadcast barring information, but will be unable to interpret the information without the schema received during the registration.

(64) At stage **812**, embodiments can communicate the barring information to the UE **110** to indicate whether the UE **110** is barred from establishing a communication session via the indicated wildcard-enabled slice group. For example, the AMF **620** broadcasts the barring information to a UE **110** including setting the one or more bits corresponding to the VNO-defined slice groups to respective barring states. Similar to the method **700** of FIG. **7**, generating and communicating the barring information in this way permits the VNO **220** to bar an entire slice group without explicitly listing all network slices that are members of the group. However, while the method **700** of FIG. **7**

is limited to barring network slice groups that can be uniquely defined as a wildcard-enabled slice group, the method **800** of FIG. **8** permits the VNO **220** full flexibility to define and bar network slice groups as desired.

(65) The remainder of the method **800** can proceed in substantially the same manner as in FIG. **7**. At stage **720**, embodiments of the UE **110** can receive and locally store the barring information. At some subsequent time, at stage **724**, the UE **110** can trigger a session establishment routine for establishing a PDU session, or other suitable communication session, between the UE **110** and the network using the network slice identified by a particular S-NSSAI **235**. As part of the routine, at stage **728**, the UE **110** can determine whether the requested S-NSSAI **235** is barred according to the barring information received and stored at stage **720** (i.e., whether the requested S-NSSAI **235** is part of a barred VNO-defined slice group). If the S-NSSAI **235** is not barred, a session can be established via the requested S-NSSAI **235** (stage **732**). If the S-NSSAI **235** is barred, a session cannot be established via the requested S-NSSAI **235** (stage **734**).

(66) The methods, systems, and devices discussed above are examples. Various configurations may omit, substitute, or add various procedures or components as appropriate. For instance, in alternative configurations, the methods may be performed in an order different from that described, and/or various stages may be added, omitted, and/or combined. Also, features described with respect to certain configurations may be combined in various other configurations. Different aspects and elements of the configurations may be combined in a similar manner. Also, technology evolves and, thus, many of the elements are examples and do not limit the scope of the disclosure or claims.

(67) Specific details are given in the description to provide a thorough understanding of example configurations (including implementations). However, configurations may be practiced without these specific details. For example, well-known circuits, processes, algorithms, structures, and techniques have been shown without unnecessary detail in order to avoid obscuring the configurations. This description provides example configurations only, and does not limit the scope, applicability, or configurations of the claims. Rather, the preceding description of the configurations will provide those skilled in the art with an enabling description for implementing described techniques. Various changes may be made in the function and arrangement of elements without departing from the spirit or scope of the disclosure.

(68) Also, configurations may be described as a process which is depicted as a flow diagram or block diagram. Although each may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may have additional steps not included in the figure. Furthermore, examples of the methods may be implemented by hardware, software, firmware, middleware, microcode, hardware description languages, or any combination thereof. When implemented in software, firmware, middleware, or microcode, the program code or code segments to perform the necessary tasks may be stored in a non-transitory computer-readable medium such as a storage medium. Processors may perform the described tasks.

(69) Having described several example configurations, various modifications, alternative constructions, and equivalents may be used without departing from the spirit of the disclosure. For example, the above elements may be components of a larger system, wherein other rules may take precedence over or otherwise modify the application of the invention. Also, a number of steps may be undertaken before, during, or after the above elements are considered.

Claims

1. A method for network slice grouping in a cellular communication network infrastructure, the method comprising: receiving a user equipment device (UE) message by at least one network function (NF) nodes of the cellular communication network infrastructure, the UE message being

received from a UE of a plurality of UE devices via a radio access network (RAN); determining, by the at least one NF nodes, that the UE message invokes a wildcard-enabled slice group that is a subset of a plurality of network slices, each network slice being a respective one of a plurality of virtualized data subnetworks, each virtualized data subnetwork uniquely identifiable by a respective one of a plurality of network slice identifiers, each network slice identifier having a respective unique combination of a slice/service type (SST) value and a slice differentiator (SD) value; generating, by the at least one NF nodes in response to the UE message, a NF message that identifies the wildcard-enabled slice group uniquely by one of the plurality of network slice identifiers that has the SD value set to designate a particular SD and has the SST value set to a predefined wildcard SST value that is one of a plurality of predefined wildcard SST values, each uniquely associated with a respective range of SST values, such that the wildcard-enabled slice group includes all network slices of the plurality of network slices that are associated with the particular SD value and are within the respective range of SST values associated with the predefined wildcard SST value; and communicating the NF message to the UE via the RAN.

2. The method of claim 1, wherein another of the plurality of network slice identifiers has the SD value set to designate a particular SD, and has the SST value set to the predefined wildcard SST value, such that the wildcard-enabled slice group includes all network slices of the plurality of network slices that are associated with the particular SD value.

3. The method of claim 1, wherein another of the plurality of network slice identifiers has the SST value set to designate a particular SST, and has the SD value set to a predefined wildcard SD value, such that the wildcard-enabled slice group includes all network slices of the plurality of network slices that are associated with the particular SST value.

4. The method of claim 3, wherein: the predefined wildcard SD value is one of a plurality of predefined wildcard SD values, each uniquely associated with a respective range of SD values; the another of the plurality of network slice identifiers has the SST value set to designate a particular SST, and has the SD value set to the predefined wildcard SD value, such that the wildcard-enabled slice group includes all network slices of the plurality of network slices that are associated with the particular SST value and are within the respective range of SD values associated with the predefined wildcard SD value.

5. The method of claim 3, wherein: the predefined wildcard SD value is one of a plurality of predefined wildcard SD values, each uniquely a designated one of the second range of SST values.

6. The method of claim 1, wherein: the UE message is a UE registration request message to register the UE with the cellular communication network, the UE registration request message indicating network slice selection assistance information (NSSAI) corresponding to a prior configuration of the UE; the determining comprises deriving from the NSSAI an allowed set of the plurality of network slices over which the UE is permitted to communicate with the cellular communication network infrastructure, the wildcard-enabled slice group representing a group of the allowed set of the plurality of network slices; and the NF message is a UE registration response message indicating to the UE the allowed set of the plurality of network slices.

7. The method of claim 1, wherein: the UE message is a session establishment request message to request establishment of a communication session between the UE and any of the wildcard-enabled slice group, the UE message including the one of the plurality of network slice identifiers that has either the SST value set to a predefined wildcard SST value or has the SD value set to a predefined wildcard SD value, thereby uniquely identifying the wildcard-enabled slice group.

8. The method of claim 1, wherein: each of the plurality of virtualized data subnetworks is associated with a stand-alone user plane function (UPF) and a stand-alone session management function (SMF); and the one or more NF nodes comprises an access and mobility management function (AMF) that is common to at least the network slices of the wildcard-enabled slice group.

9. The method of claim 1, wherein: the cellular network infrastructure operates according to a set of predefined networking standards promulgated by a standard setting organization; each of the

plurality of network slice identifiers is formatted according to a schema defined by the set of predefined networking standards, the schema reserving a first range of the SST values for definition by the standard setting organization and reserving a second range of the SST values for definition by a virtual network operator associated with the plurality of virtualized data subnetworks; and the predefined wildcard SST value is a designated one of the first range of SST values.

10. The method of claim 1, wherein: the cellular network infrastructure operates according to a set of predefined networking standards promulgated by a standard setting organization; each of the plurality of network slice identifiers is formatted according to a schema defined by the set of predefined networking standards, the schema reserving a first range of the SST values for definition by the standard setting organization and reserving a second range of the SST values for definition by a virtual network operator associated with the plurality of virtualized data subnetworks; and the predefined wildcard SST value is a designated one of the second range of SST values.

11. A system for network slice grouping in a cellular communication network infrastructure, the system comprising: a plurality of virtualized data subnetworks built on the cellular communication network infrastructure, each virtualized data subnetwork uniquely being one of a plurality of network slices identifiable by a respective one of a plurality of network slice identifiers, each network slice identifier having a respective unique combination of a slice/service type (SST) value and a slice differentiator (SD) value; and one or more network function (NF) nodes in communication with the plurality of virtualized data subnetworks, in communication with a plurality of user equipment devices (UEs) via one or more radio access networks (RANs), and comprising: one or more processors; and non-transitory memory having instructions stored thereon, which, when executed, cause the one or more processors to perform steps comprising: receiving a UE message from a UE of the plurality of UE devices via a RAN of the one or more RANs; determining that the UE message invokes a wildcard-enabled slice group that is a subset of a plurality of network slices and is uniquely identifiable by one of the plurality of network slice identifiers that has the SD value set to designate a particular SD and has the SST value set to a predefined wildcard SST value, such that the wildcard-enabled slice group includes all network slices of the plurality of network slices that are associated with the particular SD value and are within the respective range of SST values associated with the predefined wildcard SST value; and communicating, to the UE via the RAN, in response to the UE message, a NF message that includes the one of the plurality of network slice identifiers to uniquely identify the wildcard-enabled slice group.

12. The system of claim 11, wherein another of the plurality of network slice identifiers has the SST value set to designate a particular SST, and has the SD value set to the predefined wildcard SD value, such that the wildcard-enabled slice group includes all network slices of the plurality of network slices that are associated with the particular SST value.

13. The system of claim 11, wherein another of the plurality of network slice identifiers has the SD value set to designate a particular SD, and has the SST value set to the predefined wildcard SST value, such that the wildcard-enabled slice group includes all network slices of the plurality of network slices that are associated with the particular SD value.

14. The system of claim 11, wherein: the UE message is a UE registration request message to register the UE with the cellular communication network, the UE registration request message indicating network slice selection assistance information (NSSAI) corresponding to a prior configuration of the UE; the determining comprises deriving from the NSSAI an allowed set of the plurality of network slices over which the UE is permitted to communicate with the cellular communication network infrastructure, the wildcard-enabled slice group representing a group of the allowed set of the plurality of network slices; and the NF message is a UE registration response message indicating to the UE the allowed set of the plurality of network slices.

15. The system of claim 11, wherein: the UE message is a session establishment request message to request establishment of a communication session between the UE and any of the wildcard-enabled slice group, the UE message including the one of the plurality of network slice identifiers that has either the SST value set to a predefined wildcard SST value or has the SD value set to a predefined wildcard SD value, thereby uniquely identifying the wildcard-enabled slice group.
16. The system of claim 11, wherein: each of the plurality of virtualized data subnetworks is associated with a stand-alone user plane function (UPF) and a stand-alone session management function (SMF); and the one or more NF nodes comprises an access and mobility management function (AMF) that is common to at least the network slices of the wildcard-enabled slice group.
17. The system of claim 11, wherein: a predefined wildcard SD value is one of a plurality of predefined wildcard SD values, each uniquely associated with a respective range of SD values; another of the plurality of network slice identifiers has the SST value set to designate a particular SST, and has the SD value set to the predefined wildcard SD value, such that the wildcard-enabled slice group includes all network slices of the plurality of network slices that are associated with the particular SST value and are within the respective range of SD values associated with the predefined wildcard SD value.
18. The system of claim 11, wherein: the cellular network infrastructure operates according to a set of predefined networking standards promulgated by a standard setting organization; each of the plurality of network slice identifiers is formatted according to a schema defined by the set of predefined networking standards, the schema reserving a first range of the SST values for definition by the standard setting organization and reserving a second range of the SST values for definition by a virtual network operator associated with the plurality of virtualized data subnetworks; and the predefined wildcard SST value is either a designated one of the first range of SST values or a designated one of the second range of SST values.
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